

B. Tech Degree VIII Semester (Supplementary)
Examination, September 2009

EC/EB/EI 804 (C) MECHATRONICS
(2002 Scheme)

Time : 3 Hours

Maximum Marks : 100

- I. (a) What is a Numerical Control System? What are the basic components of a Numerical Control System? Explain the point to point and contouring systems. (15)
(b) What are the advantages of NC over conventional machine tools? (10)
- OR**
- II. (a) Write short notes on :
(i) Accuracy
(ii) Productivity as applied to CNC machines. (10)
(b) What are the different types of guide ways used in CNC machines? (15)
- III. (a) Explain hydraulic system drives used in CNC machines. (10)
(b) Discuss the working of DC servomotors and stepping motors. (15)
- OR**
- IV. (a) Explain the following feedback devices :
(i) Encoders (ii) Resolvers
(iii) Inductosyn (iv) Tachometers. (16)
(b) Explain incremental open-loop control system. (9)
- V. (a) Explain what is Automated Storage and Retrieval Systems (ASRS). What are the different types of ASRS? What are the different elements in ASRS? Explain. (15)
(b) What is group technology? Explain. (10)
- OR**
- VI. (a) Explain variant process planning and generative process planning. Compare these two approaches to process planning. (15)
(b) Explain machine vision. Discuss the following three functional components of vision system.
(i) Image acquisition
(ii) Processing
(iii) Output or display (10)
- VII. (a) Explain
(i) Resolution
(ii) Repeatability of robotic systems. (10)
(b) Explain the five basic configurations commonly available in commercial industrial robots. (15)
- OR**
- VIII. (a) Explain Robot language structure. Also explain the generations of Robot programming languages. (10)
(b) Explain the following methods of defining positions in space.
(i) Joints movements
(ii) XYZ coordinate motions (world coordinate)
(iii) Tool coordinate motions. (15)